

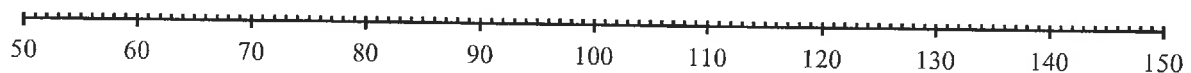
QUESTION 1

The tuck shop at Great Future High School sells cans of soft drinks. The Environmental Club at the school decided to have a can-collection project for three weeks to make learners aware of the effects of litter on the environment.

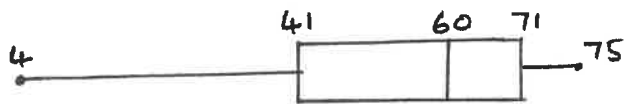
The data below shows the number of cans collected on each school day of the three-week project.

58	83	85	89	94
97	98	100	105	109
112	113	114	120	145

- 1.1 Calculate the mean number of cans collected over the three-week period. (2)
 - 1.2 Calculate the standard deviation. (2)
 - 1.3 Determine the lower and upper quartiles of the data. (2)
 - 1.4 Use the scaled line on DIAGRAM SHEET 1 to draw a box and whisker diagram to represent the data. (3)
 - 1.5 On how many days did the number of cans collected lie outside ONE standard deviation of the mean? (3)
- [12]**



QUESTION 2

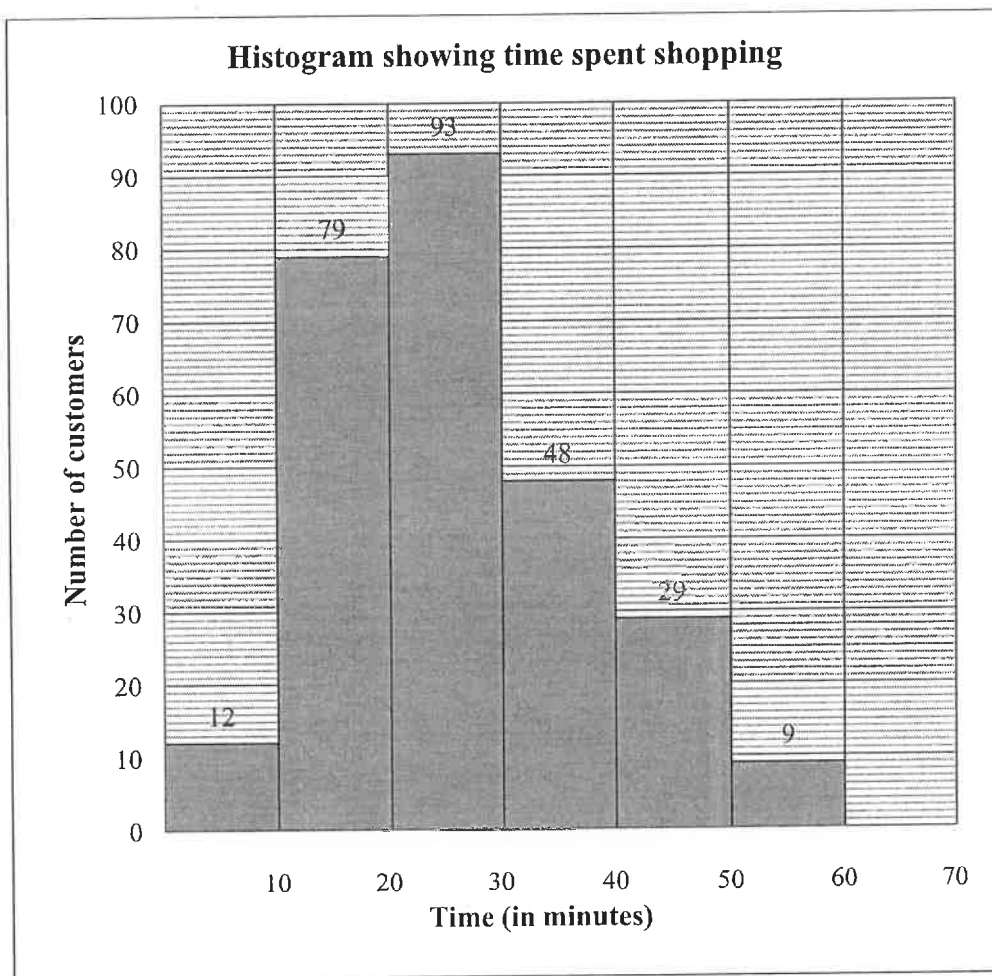


The above BOX-AND-WHISKER diagram shows how marks are distributed in a class.

- 2.1. Determine the
 - 2.1.1. range
 - 2.1.2. inter-quartile range
- 2.2. Comment on the distribution of the data.
- 2.3. Farhaan claims that the mean mark must be less than 60. Is he correct? Motivate.
- 2.4. What percentage of learners have marks greater than 41?
- 2.5. Which of the mean or median would be the best measure of central tendency in this case?

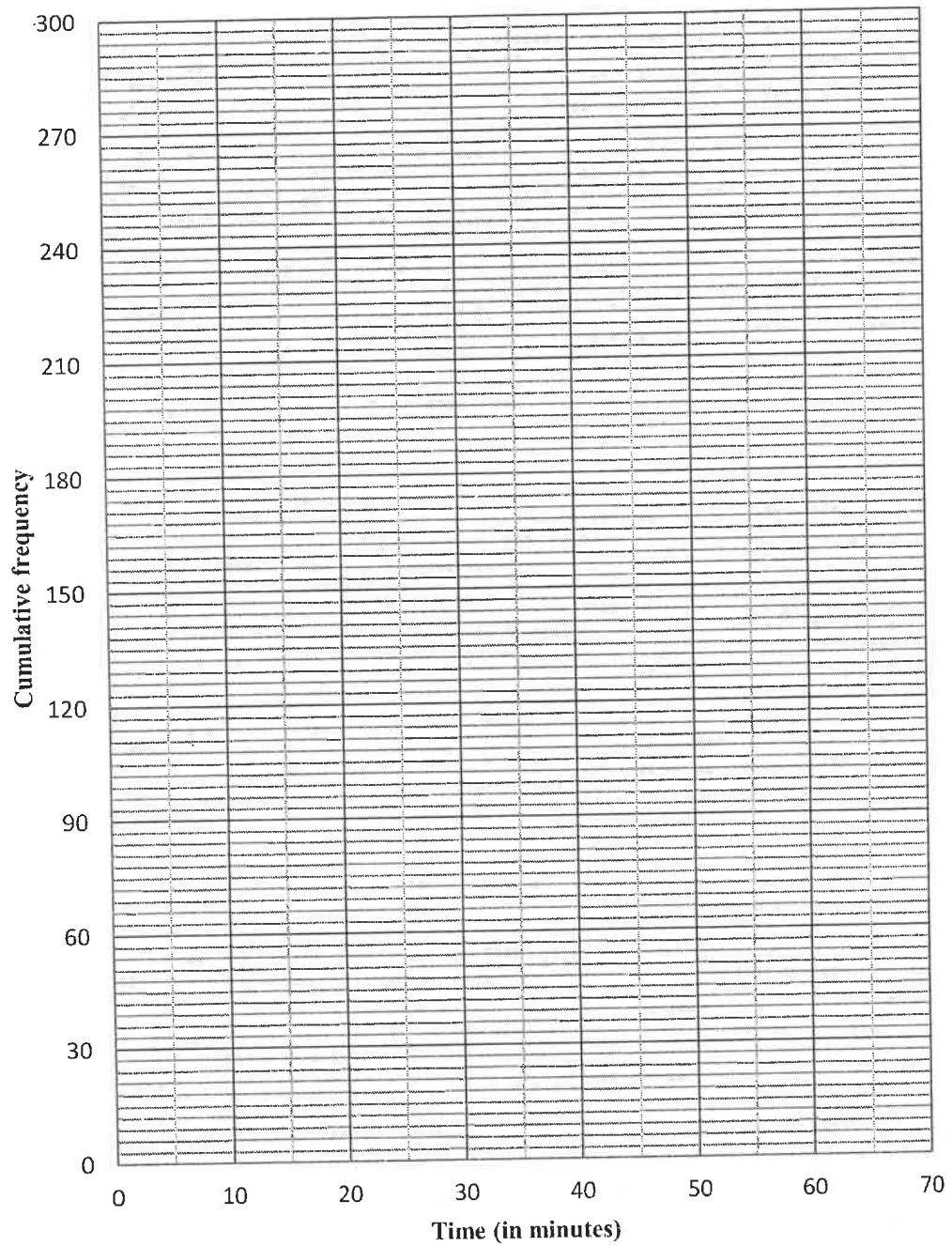
QUESTION 3

The histogram below shows the time, in minutes, spent by customers while shopping at Excellent Supermarket.



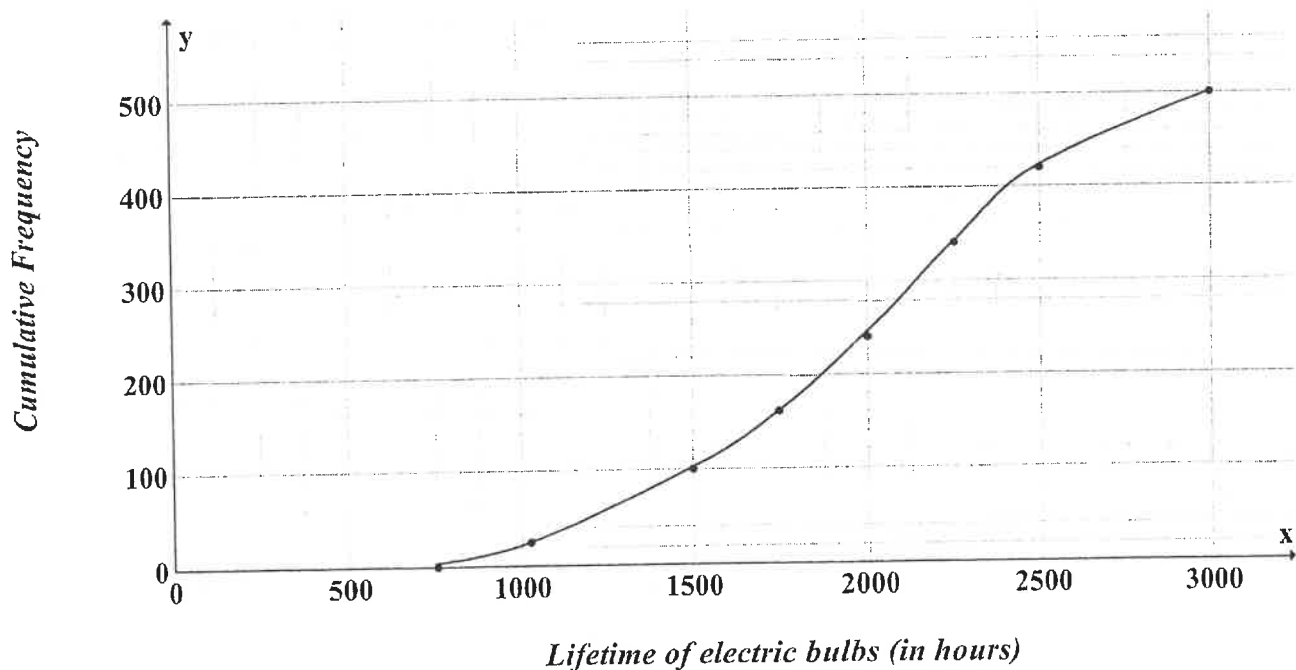
- 3.1 Complete the frequency column and cumulative frequency column in the table on DIAGRAM SHEET 1. (3)
- 3.2 Use the grid on DIAGRAM SHEET 2 to draw the ogive of the above data. (4)
- 3.3 Use the ogive to estimate the median time that customers spent at this supermarket. (2)
- 3.4 Comment on the skewness of the data. (1)

Cumulative frequency curve of time spent shopping



QUESTION 4

The lifetime of electric light bulbs was measured in a laboratory. The results are shown in the cumulative frequency curve below.



4.1 Use the above cumulative frequency curve to determine the following:

4.1.1 How many light bulbs were tested? (1)

4.1.2 The median lifetime of the electric light bulbs tested. (2)

4.1.3 The interquartile range. (2)

4.1.4 The number of electric light bulbs with a lifetime of between 1750 and 2000 hours. (2)

4.1.5 Write down the modal class. (1)

4.2 If the cost of one light bulb is R5,00, determine the amount spent on purchasing the light bulbs that lasted longer than 2 500 hours. (2)

QUESTION 5

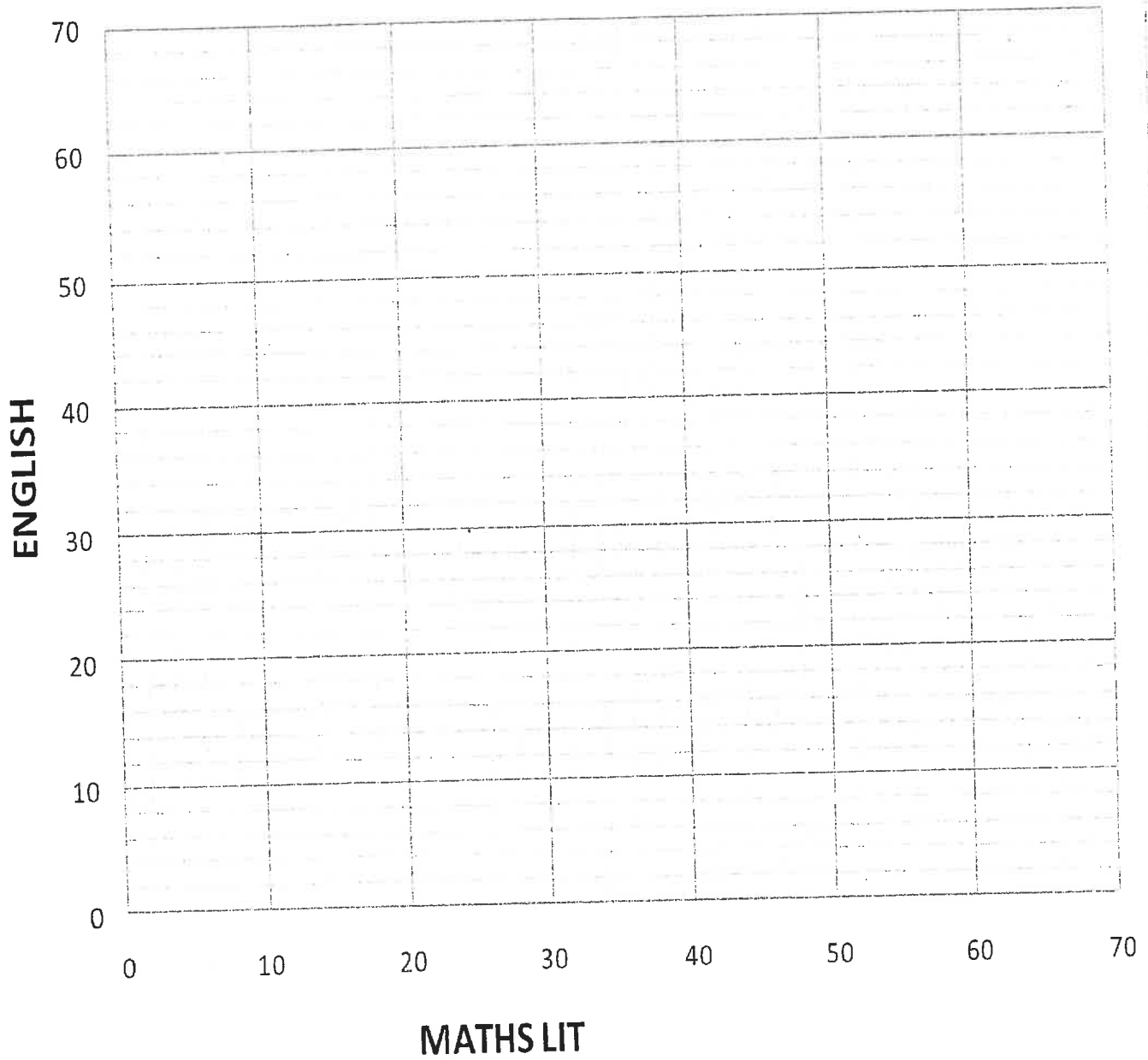
Maths Literacy teachers are always complaining about their learners' language and reading skills.

Below we have the June results for some matric learners in English and Mathematical Literacy.

MATHS LIT	25	38	40	46	47	12	49	49	54	55	59	64
ENGLISH	34	53	62	40	44	20	50	55	61	63	54	59

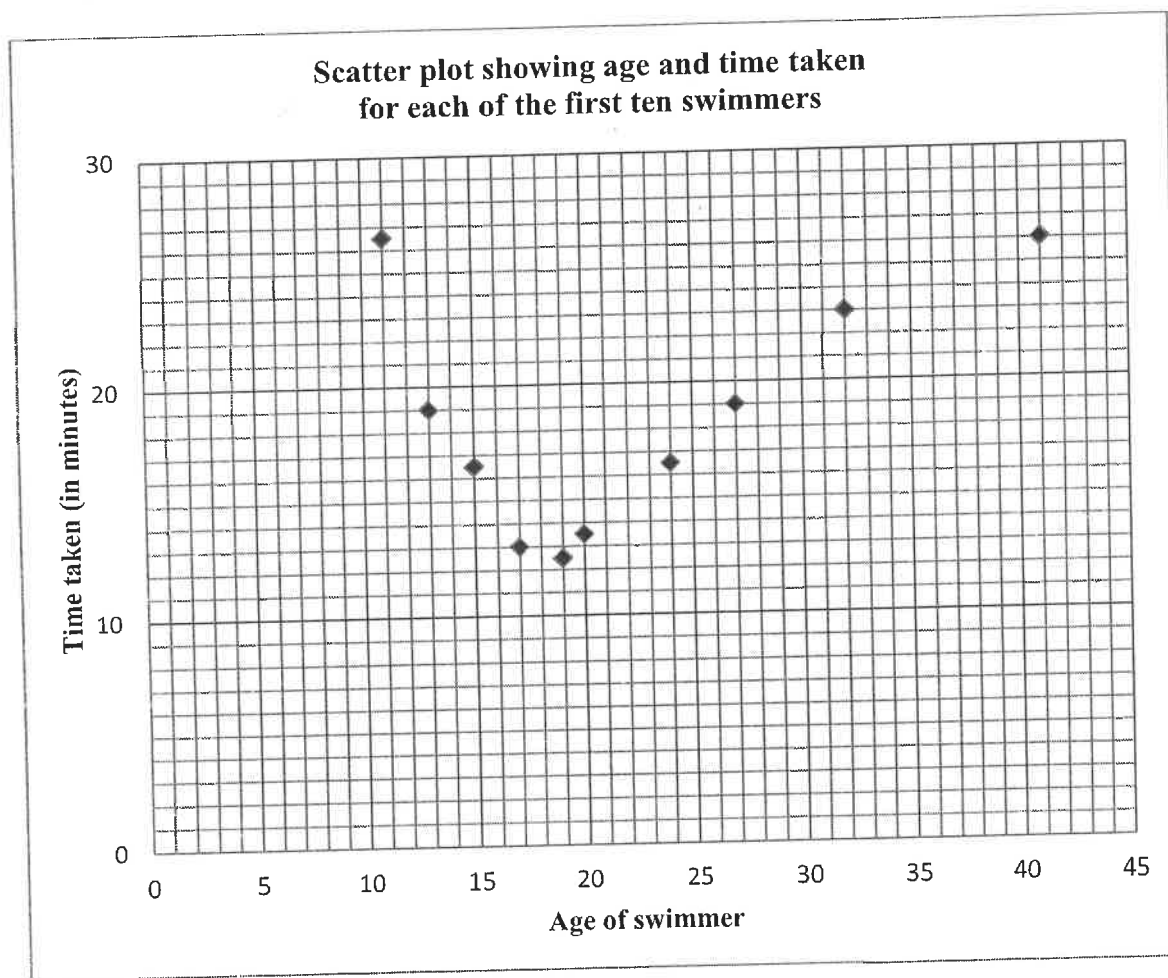
- 5.1 Use the grid provided on **DIAGRAM SHEET 1** to draw a scatter plot for the data. (3)
- 5.2 Using the least squares method, determine the equation of the regression line (4)
- 5.3 Draw the least squares line for the data on the scatter plot diagram drawn in QUESTION 1.1 (**DIAGRAM SHEET 1**). (2)
- 5.4 Estimate the Math Literacy results for a learner who gets 58% for English. (2)
- 5.5 Estimate the English results for a learner who gets 69% for Math Literacy. (2)
- 5.6 Determine the correlation coefficient between the English and Math Literacy results. (2)
- 5.7 Use the correlation coefficient to comment on the relationship between the English and Math Literacy results. (2)
- 5.8 Determine the standard deviation of the Math Literacy results. (2)

MATHS LIT VS ENGLISH



QUESTION 6

The scatter plot below shows the age and the time taken for each of the first ten swimmers of a swimming club to complete an open water swimming event. The time taken is rounded to the nearest half-minute.



- 6.1 Write down the coordinates of an outlier in the scatter plot. (1)
- 6.2 Which of the following functions will best fit the data: linear, quadratic or exponential? (1)
- 6.3 Give an explanation for the trend observed in this set of data. (2)
- 6.4 If the two worst (longest) times are disregarded from the set of data, how will this affect the following:
 - 3.4.1 The standard deviation of the original set of data (1)
 - 3.4.2 The mean of the original set of data (1)

Data Handling: Solutions

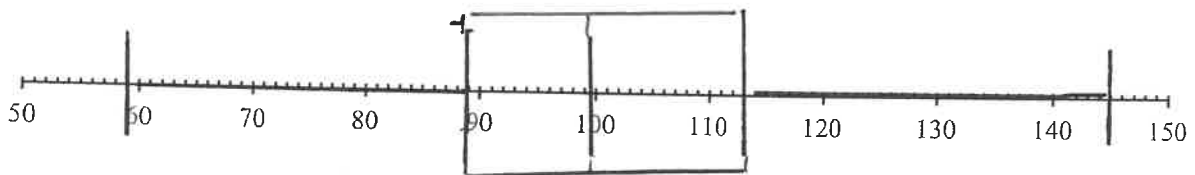
Question 1:

1.1. $101, 47$

1.2. $19, 07$

1.3. $Q_1 = 89$
 $Q_3 = 113$

1.4.



1.5 $(\mu - \sigma; \mu + \sigma)$

$= (101, 47 - 19, 07; 101, 47 + 19, 07)$

$= (82, 4; 120, 54)$

$\therefore$ On two days.

Question 2:

2.1.1. $75 - 4$

$= \underline{71},$

2.1.2. $71 - 41$

$= \underline{30},$

(2)

2.2 Data is negatively skewed. ($\text{mean} < \text{median}$)

2.3. Yes; the minimum value of 4 is very far removed from the other values and can possibly be an outlier. The extreme low value has a greater effect on the mean than the median. $\therefore \text{mean} < \text{median}$

2.4. 75%

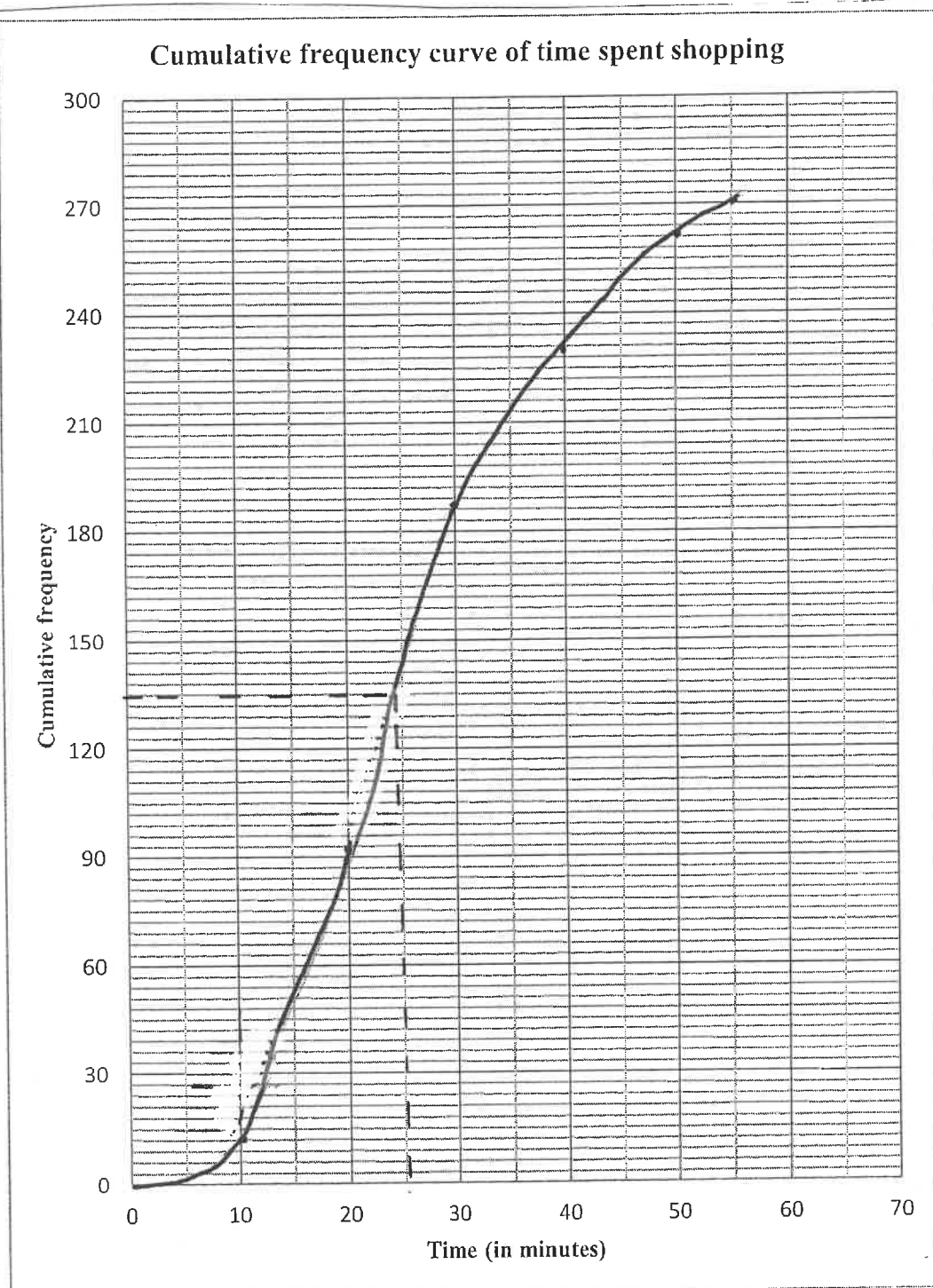
2.5. Median due to the extremely low value of 4 having a greater effect on the mean

Question 3

3.1	Class	f	cf
	$0 \leq x < 10$	12	12
	$10 \leq x < 20$	79	91
	$20 \leq x < 30$	93	184
	$30 \leq x < 40$	48	232
	$40 \leq x < 50$	29	261
	$50 \leq x < 60$	9	270

3.2.

3



3.3.

$$n = \frac{1}{2} (270)$$

$$= 135$$

$\therefore$ 25 minutes

3.4

positively skewed.

Question 4:

$$4.1.1. \quad 500$$

$$4.1.2. \quad n = \frac{1}{2} (500) \\ = 250$$

$\therefore$ Lifetime of 2000 hours.

$$4.1.3 \quad Q_1 \quad \text{position} = \frac{1}{4} (500) \\ = 125$$

$$Q_1 = 1625$$

$$Q_3: \quad \text{position} = \frac{3}{4} (500) \\ = 375$$

$$Q_3 = 2325$$

$$IQR = 2325 - 1625 \\ = \underline{700},$$

$$4.1.4 \quad 240 - 160 = \underline{80},$$

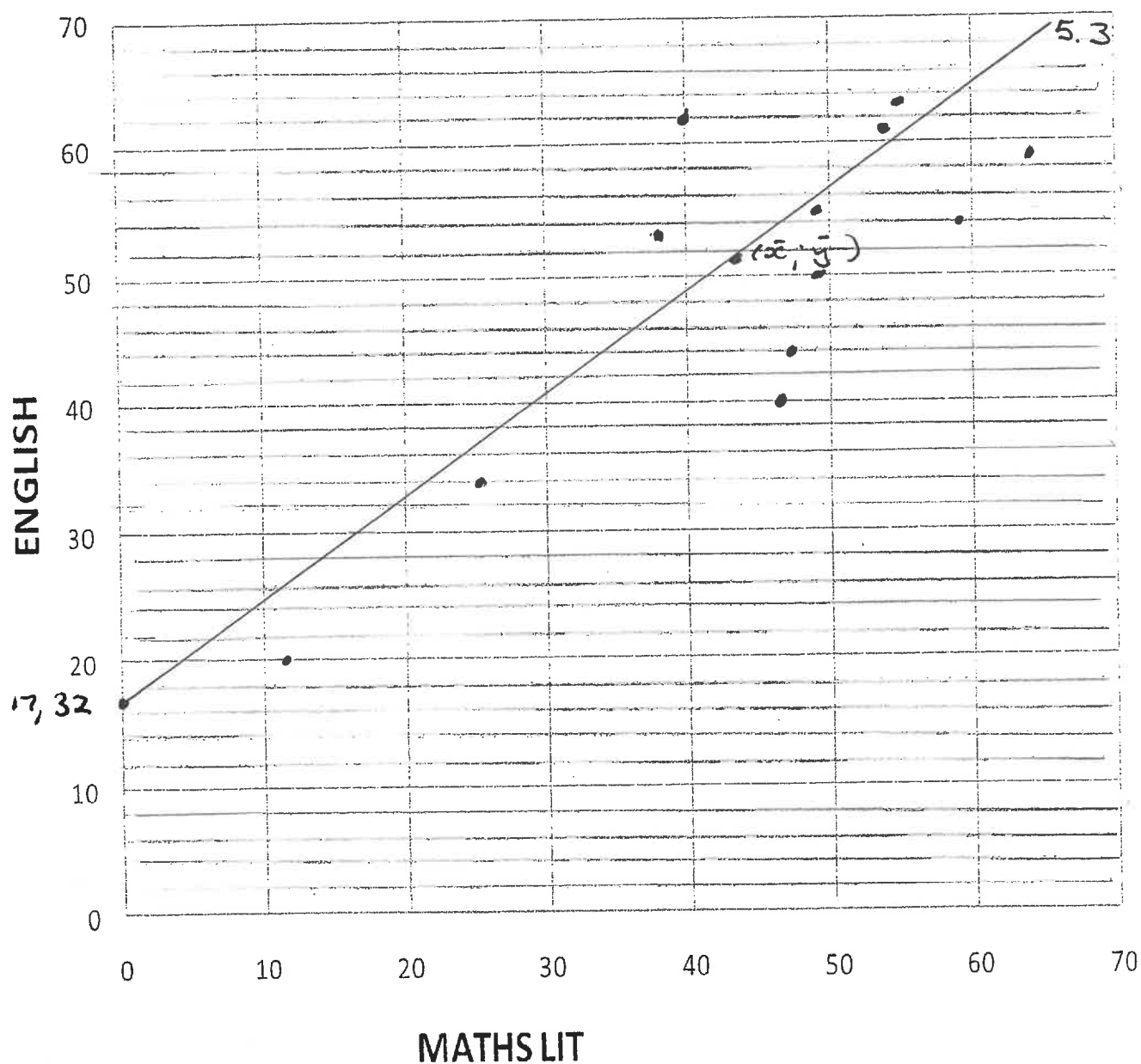
$$4.1.5 \quad 2000 - 2250$$

$$4.2. \quad (1500 - 420) \times 5 \\ = \underline{R490},$$

Question 5 :

5.1.

MATHS LIT VS ENGLISH



5.2 $y = 0,72x + 17,32$

5.3. $\bar{x} = 44,83$
 $\bar{y} = 49,58$

5.4 $y = 0,72x + 17,32$
 $58 = 0,72x + 17,32 \Rightarrow x = 56,5$
 $\approx 57\%$

5.5 $y = 0,72(69) + 17,32$
 $= \underline{67,}$

5.6 $r = \underline{0,81},$

5.7 Strong positive Linear Correlation

5.8 $\sigma = \underline{13,96},$

Question 6

6.1 (41; 26)

6.2 Quadratic

6.3 Swimming times decrease until the peak age (± 18 years). After that times start to increase, as people age.

6.4 smaller standard deviation

6.5 lower mean